SCIENTIFIC COMMENTARY

Stem cell-derived dopamine cell therapies for Parkinson's disease: what have the first trials shown?

This scientific commentary refers to 'Phase I/II trial of iPS-cell-derived dopaminergic cells for Parkinson's disease' by Sawamoto *et al.* (https://doi.org/10.1038/s41586-025-08700-0) and 'Phase I trial of hES cell-derived dopaminergic neurons for Parkinson's disease' by Tabar *et al.* (https://doi.org/10.1038/s41586-025-08845-y).

Parkinson's disease (PD) is a common neurodegenerative disorder that has as part of its core pathology, the loss of the A9 nigral dopaminergic (DAergic) neurons which project to the striatum, particularly the posterior putamen. This leads to reduced dopamine levels and a range of motor features. Replacement of this DAergic input using drug therapies has been the mainstay of treatment for PD for over 50 years but brings with it long-term complications including motor fluctuations and neuropsychiatric problems. Managing these typically requires the use of increasingly complex drug regimes, with many patients eventually needing more advanced interventions such as invasive DAergic pump therapies or neurosurgery to control their condition.

Given these challenges, a longstanding ambition of the field has been to repair the DAergic network by transplanting DAergic precursor cells to replace those lost to the disease process. This pioneering approach began in the 1980s in Lund, Sweden, using human fetal ventral mesencephalic (hfVM) tissue and has continued to the present day with the recent TransEuro trial, which sought to refine this method.[1] As well as confirming earlier findings of highly variable outcomes,[2] the trial highlighted logistical issues—including inconsistent tissue supply and difficulties in standardising the final cell suspension—which continue to limit the feasibility of the approach.[3]

Over the last 15 years, efforts have been made to address these problems through the use of pluripotent stem cell-derived dopamine precursor cells, which have now entered first-in-human

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trials.[4,5] There are currently two main sources of human pluripotent stem cells for clinical use: human embryonic stem cells (ESCs), derived from the inner cell mass of pre-implantation blastocysts, and induced pluripotent stem cells (iPSCs), derived from reprogrammed adult somatic cells. Both can be differentiated into authentic, functional dopaminergic neurons suitable for transplantation[6] and much work has been done to qualify such cells for clinical trials.[7,8]

The first clinical trials using pluripotent stem cells were initiated in 2018 in Kyoto, Japan, using allogeneic iPSC-derived dopaminergic cells, and in 2019 in New York using an allogeneic ESC-derived therapy. The outcomes of both trials were recently published in *Nature*.[4,5]

In the phase I trial conducted by BlueRock Therapeutics in New York, 12 patients with moderately advanced PD received bilateral transplants into the putamen of a cryopreserved allogeneic dopaminergic progenitor cell product known as bemdaneprocel.[5] Patients were administered one of two doses (the first 5 patients received a lower dose than the last 7) and all patients received a year of immunosuppression, similar to that used in organ transplants (basiliximab, steroids and tacrolimus) (Fig. 1A). No major product-related adverse events were reported, indicating that the cells and transplant procedure were well tolerated, including this level of immunosuppression in patients with this degree of PD. A modest clinical response was observed in the high-dose group, although this was not accompanied by a significant change on ^{18}F-DOPA PET imaging or reductions in PD medication doses.

In the phase I/II trial at Kyoto University, 7 patients with moderately advanced PD were transplanted with allogeneic iPSC-derived dopaminergic precursor cells, freshly manufactured for each patient and delivered bilaterally into the posterior putamen as small aggregate spheres.[4] Three patients received a lower dose and four a higher dose with some of the recipients being HLA matched to the implanted tissue. All patients received 1 year of tacrolimus immunosuppression post-surgery (Fig. 1B).

No serious adverse events were reported up to 2 years post-grafting, and MRI scans showed no evidence of graft overgrowth, supporting the safety and feasibility of the procedure (the primary outcome measure). The secondary endpoints focused on clinical efficacy and dopaminergic PET scans, with modest changes observed. Some of the patients improved on clinical measures and ^{18}F-DOPA scans provided evidence of graft survival, which is encouraging, but in all cases effect sizes were modest and variable.

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So, what can we take away from these two trials? The question is particularly apposite as BlueRock Therapeutics are now moving towards a Phase III trial—exPDite-2—a randomized, double-blind, sham surgery-controlled trial to assess the efficacy and safety of bemdaneprocel in people with moderate PD, due to start this year. If successful, this trial could lead to the therapy gaining regulatory approval. In addition, the Japanese team can now apply for conditional approval, based on the findings from their first trial, which would allow for early market access under the Pharmaceuticals and Medical Devices Act in Japan.

Firstly, we can conclude that trials using human pluripotent stem-cell derived dopaminergic precursors are safe. Mass lesions have not been observed post-grafting and the surgical procedure itself appears to have been well-tolerated, albeit in the small number of patients that have been grafted to date. In addition, the immunosuppressive regimes in these trials seem to have been tolerated without major adverse effects, including after discontinuation.

Secondly, it appears that none of the patients have shown a dramatic reversal of their PD in terms of clinical response, change in PD medication requirements, or ^{18}F-DOPA PET imaging. In the bemdaneprocel trial, a statistically significant clinical response was observed in the higher dose cohort but was not matched by changes in dopaminergic signal on imaging, which one might have expected had the benefits been due to cell survival and increased dopaminergic levels. This disconnect suggests that the grafts may not contain enough surviving dop/aminergic cells to innervate the putamen and restore the putamenal dopaminergic signal—in contrast to what has been seen in some patients with hfVM transplants.[9,10] The apparently low survival rates of dopaminergic neurons at the site of grafting may relate to:

(i) Suboptimal dosing: Cell numbers were derived from calculations based on preclinical animal models, and may have been too low for human patients

(ii) Cell loss during preparation and surgical delivery: There is currently no generally accepted and proven device for delivering DAergic cells to the human brain

(iii) Loss of cells post grafting: Cells may be lost due to the host environment and/or immune rejection

(iv) Failure of differentiation: Grafted progenitors may fail to differentiate into mature innervating midbrain DAergic cells in the human PD brain.

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To resolve these unknowns, further iterative, small-scale trials are needed prior to advancing to phase III trials. These smaller studies should seek to optimise the dose, delivery method, and optimal procedures to support and differentiate the grafted cells. Critically, we must identify the dose of DAergic progenitors that can reliably restore putamenal dopamine levels to normal with matching clinical improvement, as has been seen in the 'best-case' hfVM grafted patients.[10] Only by achieving this can we establish cell-based therapies as a viable, competitive treatment option for people with Parkinson's disease.

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Competing interests

M.P. is the owner of Parmar Cells that holds related IP (U.S. patent 15/093,927, PCT/EP17181588) and provides paid consultancy advice to Novo Nordisk. R.A.B. has provided

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paid consultancy advice to Novo Nordisk, BlueRock Therapeutics/Bayer and Aspen Neuroscience.

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Figure legend

Figure 1 Schematic showing the approach and trial design for the two recently published stem cell-derived dopaminergic cell transplant studies in patients with Parkinson's disease. (**A**) Tabar *et al.*[5] (**B**) Sawamoto *et al.*[4] Created in BioRender. Sebestikova, J. (2025) https://BioRender.com/m2u1ttx.

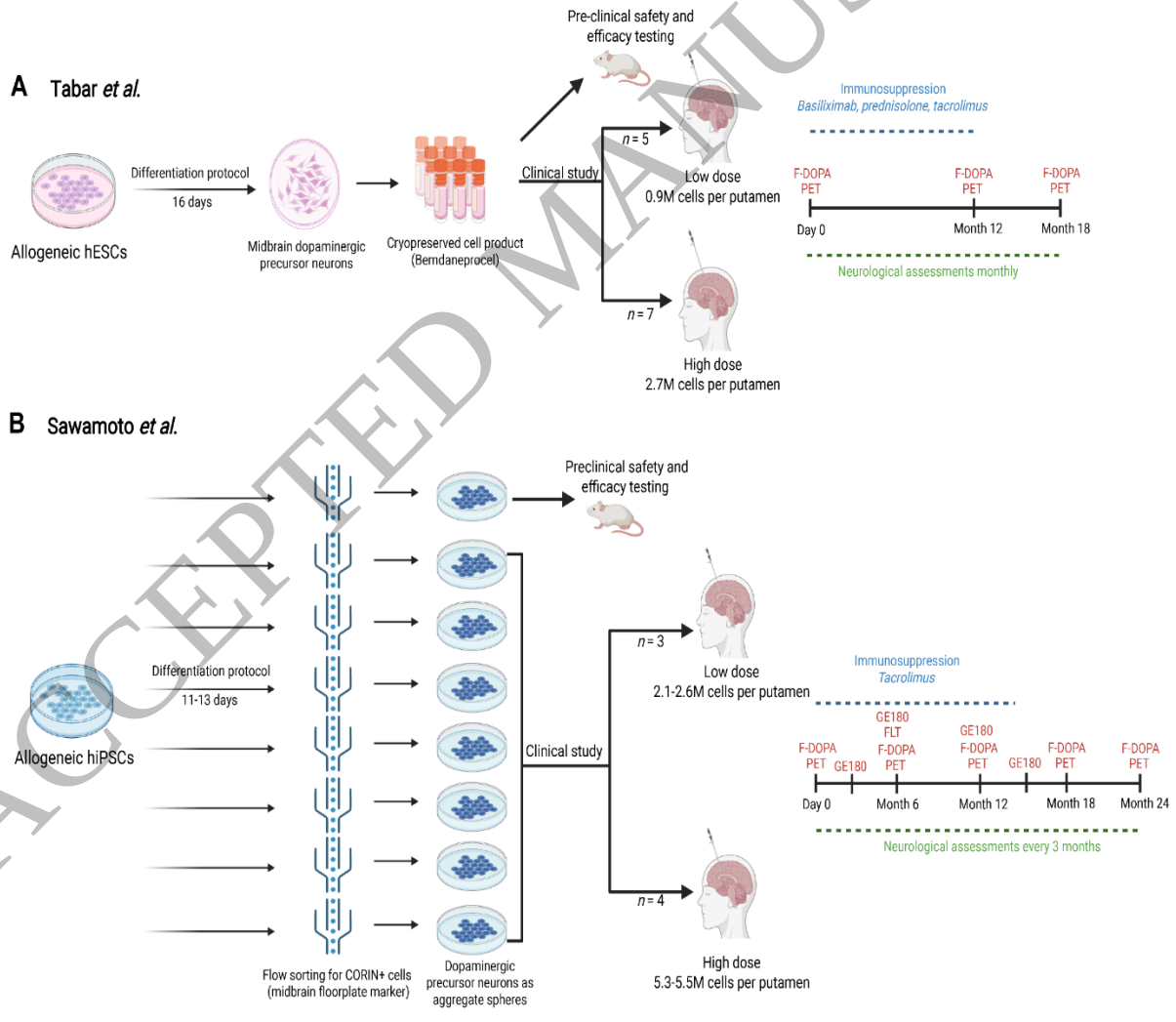

Figure 1
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